Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to present the institute's proprietary technologies and identify potential areas of collaboration with Korean SMEs, enabling joint research and development under the 2026 TIPA International Joint R&D Program.
This proposal aims to demonstrate the technological readiness, industrial relevance, and scalability of the concepts, providing a clear basis for matching with SME partners.
The Technology Proposal will be publicly disclosed alongside the official announcement of the 2026 TIPA International Joint R&D Program.

Management Summary

This Technology Proposal addresses the development of a **modular electric motorcycle platform** tailored to urban mobility applications and **SME-driven manufacturing environments**. The concept is based on a configurable construction kit that enables **customer-specific frame design** while maintaining a standardized and scalable platform architecture suitable for small-series production.

At the core of the platform is a **modular structural frame** manufactured from sustainable, separable, and recyclable materials, with a strong emphasis on wood-based composite materials in the current development stage. This structural approach supports resource efficiency, circular economy principles, and flexible customization while preserving industrial feasibility. The frame functions as a configurable backbone that can be adapted to different user requirements and vehicle variants without requiring a complete redesign.

The platform integrates a combination of **newly developed sustainable components**—such as seating systems, ergonomic interfaces, and storage elements—with **proven, standardized industrial components**, including wheels, braking systems, and suspension elements. This hybrid integration strategy ensures technical reliability, cost efficiency, and compatibility with established supply chains.

A central element of the proposed joint R&D project is the integration of a **modular electric drivetrain**, enabling locally emission-free operation and suitability for dense urban environments where reduced noise, emissions, and space efficiency are critical. The drivetrain architecture is designed for scalability and compatibility across multiple vehicle configurations.

By combining electrification, modularity, and sustainable component reuse, the proposed platform creates **new business opportunities for Korean SMEs**, particularly in retrofit solutions, customization, and small-series electric mobility production. Through international collaboration under the **2026 TIPA International Joint R&D Program**, the project aims to mature the concept into an **industrially applicable technology platform** aligned with Korea's strategic priorities in advanced mobility, sustainability, and SME innovation.

In addition, the platform follows a **design-for-compliance approach**, leveraging low-voltage (48V) electrification and the systematic reuse of certified components to reduce regulatory complexity and support future vehicle approval processes and market deployment in urban mobility contexts.

1. Technology Proposer Information

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2. Technology Overview

<table>
<tr><td>Title of Proposed Technology</td><td colspan="4">Modular Electric Motorcycle Retrofit Platform for Sustainable Urban Mobility</td></tr>
<tr><td>Korea's 12 National Strategic Technology Fields</td><td colspan="4">Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (X), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )</td></tr>
<tr><td rowspan="4">Industrial Technology Classification [1)]</td><td colspan="2">Technology Field 1 (Large Category)</td><td colspan="2">Technology Field 2 (Large Category)</td></tr>
<tr><td colspan="2">Mechanical Materials</td><td colspan="2">Mechanical Materials</td></tr>
<tr><td>Medium Category</td><td>Small Category</td><td>Medium Category</td><td>Small Category</td></tr>
<tr><td>Automobile / Rail Vehicles</td><td>Vehicle Body and Lightweight technologies</td><td>Automobile / Rail Vehicles</td><td>Electrical and Electronic Systems</td></tr>
<tr><td>Concept of Proposed Technology</td><td colspan="4">The proposed technology is a modular retrofit platform for the electrification and sustainable reuse of existing motorcycles, specifically targeting urban mobility applications and SME-driven manufacturing environments. Unlike conventional approaches that focus on the development of entirely new electric vehicles, the project emphasizes the systematic reuse of high-quality, existing motorcycle components, which are integrated into a newly developed modular platform architecture. This strategy significantly reduces material consumption, development effort, and environmental impact while maintaining high functional, ergonomic, and safety standards.

At the core of the proposed technology is a modular structural and electrical system architecture that enables the integration of a 48V electric powertrain across different motorcycle configurations. Standardized mechanical, electrical, and functional interfaces provide a high degree of flexibility in component selection, customization, and scalability across multiple vehicle classes. This modularity supports ergonomic adaptation, functional variation, and small-series or variant-oriented production, without requiring a complete redesign of the vehicle.

From a regulatory and market-readiness perspective, the platform follows a design-for-compliance strategy aligned with relevant light vehicle and L-category motorcycle frameworks. By prioritizing a low-voltage (48V) electric architecture and the reuse of industrially proven, certified components for safety-critical systems—such as brakes, suspension, and wheels—the platform aims to reduce homologation complexity and approval risks. The separation of structural, functional, and electrical subsystems further facilitates regulatory adaptation across different vehicle variants and target markets.

The proposed technology comprises the following main functional elements:

• Modular frame and mounting architecture compatible with reused components such as wheels, suspension systems, braking systems, and selected chassis elements.
• 48V electric drivetrain integration optimized for urban mobility use cases, balancing performance, safety, and regulatory compatibility.
• Retrofit-oriented integration process, including disassembly, component evaluation, system integration, and functional validation.
• Design principles supporting SME manufacturability, such as simplified assembly processes, material efficiency, and scalable production concepts.
• Urban mobility and sustainability benefits, including reduced traffic space usage during operation and parking, as well as locally emission-free operation enabled by the electric drivetrain.

An initial prototype demonstrating the fundamental feasibility of the concept has already been developed and publicly presented (cf. Figure 1). Within the scope of the proposed project, the experience gained from this prototype will be abstracted and generalized into a scalable technology platform, enabling the transition from a single demonstrator toward an industrially applicable retrofit system.</td></tr>
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Figure 1: Functional prototype of modular electric retrofit motorcycle integrating high-value reused components and 48V drivetrain.

Potential applications of the proposed technology include:

<ul>
<li>Urban electric motorcycles for sustainable city transport.</li>
<li>Retrofit solutions for existing vehicles, promoting circular economy principles.</li>
<li>New SME-driven business models in customization, small-series production, and electrified mobility solutions.</li>
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<td>Rationale for Joint R&D Project</td>
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The rapid pace of urbanization and the global transition toward electric mobility are driving increasing demand for sustainable and resource-efficient transportation solutions. In the two-wheeler sector, however, current electrification strategies predominantly focus on the development of entirely new vehicles. These approaches typically require high material input, substantial upfront investment, and long development cycles, limiting accessibility for small and medium-sized enterprises (SMEs) and constraining the overall sustainability impact.

The proposed Joint R&D project addresses this gap by pursuing a retrofit-based electrification strategy combined with the circular reuse of existing vehicle components. While offering significant sustainability and cost advantages, this approach introduces complex technical and organizational challenges. These include modular system architecture design, electric drivetrain integration into existing vehicle structures, structural and ergonomic optimization, sustainability-driven material selection, and the transition from prototype concepts to industrially scalable solutions. Such challenges cannot be effectively addressed by a single organization acting independently.

International collaboration is therefore a core requirement for the successful realization of the proposed technology. Hochschule Esslingen contributes expertise in modular system design, sustainability-oriented engineering methodologies, and applied research, supported by hands-on experience gained from the development and public demonstration of an initial functional prototype. Korean SME partners contribute complementary strengths in industrial manufacturing, component development, testing and validation, supply-chain integration, and commercialization—capabilities that are essential for transforming a technical concept into a market-ready and scalable product platform.

Korea provides a particularly suitable environment for this collaboration due to its strong SME-based manufacturing ecosystem, its strategic national focus on advanced, electric, and sustainable mobility, and its policy-driven support for SME-led innovation. Through joint R&D activities, international system and sustainability knowledge can be effectively transferred into the Korean industrial context, while Korean SMEs gain access to novel platform concepts and expand their technological and business capabilities in electric and circular mobility markets.

The Joint R&D framework provided by TIPA is therefore essential to:

<ul>
<li>Reduce technical and commercialization risks through shared development and validation.</li>
<li>Align technology development with industrial manufacturing constraints and market requirements.</li>
<li>Accelerate system integration, scaling, and market entry.</li>
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	• Strengthen the international competitiveness of Korean SMEs in emerging electric mobility sectors. This collaborative approach ensures that the proposed technology evolves from a validated prototype concept into an **industrially viable, scalable, and commercially relevant retrofit platform**, delivering measurable technological, environmental, and economic impact.
Key Words	1. Modular Retrofit Platform 2. Urban Electric Motorcycle 3. Circular Component Reuse 4. 48V Low-Voltage Powertrain 5. SME-Manufacturable System

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	**TRL 4** – Technology validated in laboratory environment, based on a functional prototype demonstrating core components and modular concept.	**End of the Project**	**TRL 6 –** Technology demonstrated in a relevant environment with integrated electric drivetrain and SME-ready modular platform for retrofit applications.
R&D Objectives	The overall objective of this Joint R&D project is to **develop and validate a scalable, SME-compatible modular retrofit platform** for the electrification and sustainable reuse of urban motorcycles, advancing from the existing prototype toward an **industrial-grade system**. To achieve this overall objective, the project pursues the following specific R&D objectives: 1. **Development of a modular retrofit system architecture** Define and validate a modular system architecture that enables integration of electric powertrains into existing motorcycles while retaining high-value components. Standardized mechanical, electrical, and functional interfaces will ensure flexibility across vehicle configurations and use cases. 2. **Integration and optimization of a 48V electric drivetrain** Develop and optimize a 48V drivetrain concept tailored for urban mobility, balancing performance, efficiency, safety, and regulatory compliance. Key tasks include powertrain packaging, thermal management, power distribution, and system-level integration within the modular platform. 3. **Development of structural and material concepts for SME manufacturing** Create adaptable, sustainable, and cost-efficient structural concepts for the modular platform. The R&D will explore **material-agnostic and hybrid approaches**, enabling Korean SMEs to implement solutions according to their manufacturing capabilities and supply chains. 4. **Definition of standardized retrofit and integration processes** Establish repeatable workflows for disassembly, component evaluation, integration, and validation. This will reduce complexity, shorten assembly time, and ensure consistent quality, making the platform suitable for **small-series production by SMEs**. 5. **Validation of performance, safety, and sustainability targets** Conduct functional testing, performance evaluation, and preliminary safety and durability assessments. Evaluate sustainability metrics such as component reuse rate and CO_2 reduction potential to demonstrate the environmental benefits of the retrofit approach. 6. **Preparation for industrialization and commercialization by Korean SMEs** Assess manufacturability, analyze cost structures, and define a commercialization roadmap tailored to Korean SME partners. This ensures smooth technology transfer and adoption in the **Korean urban mobility market**.			

	By addressing these objectives, the project bridges the gap between a **prototype-level concept** and an **industrially viable, scalable, and commercially relevant modular retrofit platform**, aligned with Korea's strategic priorities in **advanced, sustainable, and SME-driven mobility**.
R&D Target Specifications	The R&D target specifications of this project are defined to ensure that the developed modular retrofit platform meets **technical feasibility, safety requirements, sustainability objectives, and SME implementation constraints**. The specifications are intended to be ambitious yet realistic, reflecting the transition from prototype-level validation toward industrial applicability. From an electrical and drivetrain perspective, the project targets a **48V electric system architecture**, which offers an optimal balance between performance, safety, regulatory compatibility, and cost-efficiency for urban mobility applications. The electric drivetrain is designed to deliver a **continuous power output in the range of 6–10 kW**, suitable for urban traffic conditions, while maintaining high energy efficiency. The target urban operating range is **60–120 km per charge**, depending on configuration and battery capacity. With respect to structural and system design, the modular platform aims to achieve a **weight reduction of at least 20%** compared to full vehicle replacement concepts, primarily through component reuse and modularization. A key performance indicator is the **reuse ratio of existing components**, with a target of **40–50% or higher**, focusing on high-value elements such as suspension systems, wheels, and braking components. Manufacturability and SME implementation are addressed through process-related targets. The project aims to define a retrofit and assembly process with a **target assembly time of less than 20 working hours per vehicle**, enabling economically viable small-series production and customization by SMEs. The modular architecture is designed to support variant flexibility without major redesign, reducing development and tooling costs. From a sustainability perspective, the project targets a **CO_2 emission reduction potential of at least 30%** compared to conventional new vehicle production, based on preliminary life-cycle considerations. Additional sustainability indicators include reduced material input and extended component life cycles through reuse. Finally, the developed system is intended to meet applicable **functional safety and regulatory requirements** for urban electric two-wheelers, providing a solid basis for further certification and market introduction by Korean SMEs.
R&D Plan	The R&D plan for the proposed modular electric retrofit platform is structured into five sequential phases, following a systematic, risk-aware development approach. Each phase builds upon the results of the previous one and incorporates clearly defined **milestones** and **termination criteria** to ensure technical feasibility, industrial relevance, and progress toward commercialization. The phased approach guides the project from the existing prototype toward a validated, SME-ready system optimized for urban mobility, sustainability, and TIPA strategic priorities. **Phase 1 – System Definition and Requirement Analysis (Months 1–3)** The project begins by establishing a comprehensive system definition for the modular retrofit platform. This includes analyzing urban mobility use cases, defining target vehicle classes, and identifying relevant regulatory, safety, and SME manufacturing requirements for 48V electric two-wheelers. Functional requirements for the modular architecture, component interfaces, and retrofit process are derived to ensure flexibility, scalability, and industrial applicability. **Milestone:** Validated system requirements and platform architecture specification. **Termination Criterion:** If critical system requirements cannot be defined to support modular integration and SME manufacturability, the project will be discontinued, as further phases depend on a robust platform foundation. **Phase 2 – Modular Platform and Interface Development (Months 4–8)** Based on the validated requirements, the modular structural and electrical platform is developed. Key

activities include the design of standardized interfaces for reused components, drivetrain integration concepts, and evaluation of material and structural solutions suitable for SME manufacturing. Simulation-based analyses and feasibility studies reduce development risks, optimize structural performance, and support ergonomically and industrially viable design decisions (cp. Figure 2). **Milestone:** Completed modular platform design and interface definitions.

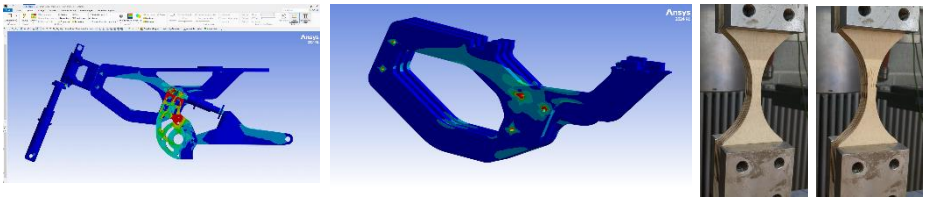

Figure 2: Conceptual approaches for modular structural frame design and load bearing analysis.

Phase 3 – Electric Drivetrain Integration and Prototype Development (Months 9–14)
This phase addresses the integration of the 48V electric drivetrain into the modular platform. Prototypical components are manufactured or sourced, and system integration follows the defined retrofit process. Functional prototypes demonstrate the feasibility of modular electrification across representative vehicle configurations. Regulatory and safety considerations are incorporated, including structural load paths, interface definitions for safety-critical components, and compliance-oriented design constraints. While full vehicle homologation is beyond the scope of this project, the activities establish a technically and structurally sound basis for future approval processes and industrial deployment (cp. Figure 3). **Milestone:** Functional prototype(s) demonstrating the integrated modular retrofit system. **Termination Criterion:** If the integrated prototypes fail to meet minimal performance, safety, or structural requirements, the project cannot proceed to validation or industrialization.

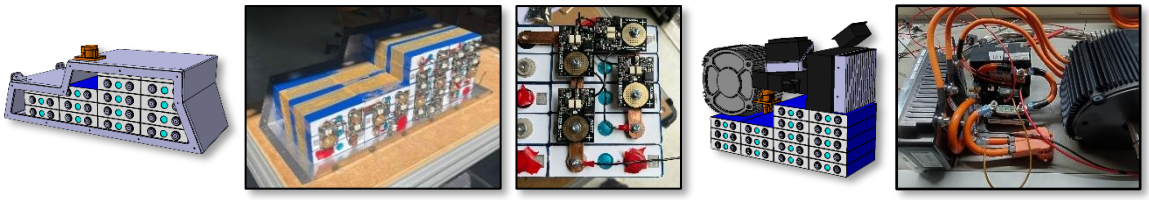

Figure 3: Integration of 48V electric drivetrain into modular prototype frame – demonstration of retrofit feasibility.

Phase 4 – Testing, Validation, and Optimization (Months 15–20)
Developed prototypes undergo systematic testing and validation, covering functional performance, safety-relevant aspects, durability, and sustainability. Testing includes verification of CO_2 reduction potential, component reuse rates, ergonomics, and operational suitability for dense urban environments. Results are used to optimize system design, assembly processes, and component selection. Particular emphasis is placed on SME manufacturability, assembly efficiency, and cost-effectiveness, ensuring readiness for small-series production. **Milestone:** Validated system performance and optimized retrofit process.

Phase 5 – Industrialization and Commercialization Preparation (Months 21–24)
The final phase prepares the platform for industrial adoption by Korean SMEs. Activities include defining manufacturing and assembly workflows, assessing scalability and cost-efficiency, and developing a commercialization roadmap aligned with SME business models. Documentation and knowledge transfer ensure a smooth transition from R&D to market-oriented implementation. **Milestone:** Industrialization-ready technology package and commercialization roadmap.

Through this phased plan, the project balances **technical development, prototype validation, regulatory preparedness, and industrial readiness**, enabling the transition from a **demonstrator-level concept to a scalable, commercially viable modular retrofit platform**. This approach ensures alignment with **Korean strategic priorities in advanced mobility, electrification, circular economy, and SME-driven innovation**, while promoting sustainable urban transportation solutions.

4. R&R (Roles and Responsibilities)

The Joint R&D project is based on a complementary division of roles between Hochschule Esslingen and the Korean SME partner. Hochschule Esslingen leads system design and technology development, while the SME partner focuses on industrial implementation, validation, and commercialization. This role distribution ensures effective collaboration, knowledge transfer, and alignment between technological innovation and industrial applicability.

Expected R&R of Research Institute	Hochschule Esslingen will assume the leading role in the system-level development and coordination of the proposed Joint R&D project. Its responsibilities focus on the conceptual, technical, and methodological foundations required to transform the existing prototype concept into a scalable, SME-ready retrofit platform. Specifically, Hochschule Esslingen is expected to: • Lead the system-level design and definition of the modular retrofit platform architecture, ensuring scalability, flexibility, and industrial applicability. • Develop structural concepts and standardized mechanical, electrical, and functional interfaces enabling the integration of reused motorcycle components and a 48V electric drivetrain. • Conduct simulation-based analyses and virtual validation, including structural integrity, load paths, ergonomics, and energy efficiency, to support design decisions and reduce development risks. • Define standardized retrofit, integration, and assembly processes tailored to SME manufacturing environments and small-series production. • Provide technical documentation, design guidelines, and structured knowledge transfer to participating SME partners, including training activities where appropriate. • Support functional testing, prototype validation, and iterative optimization with respect to performance, safety-related aspects, and sustainability targets. • Coordinate international collaboration and contribute to overall project management, ensuring alignment across partners and timely achievement of defined milestones. Through these contributions, Hochschule Esslingen will ensure technical coherence, methodological rigor, and effective knowledge transfer, forming the backbone of the joint development effort and enabling successful industrial adoption by Korean SMEs.
Expected R&R of Korean SME Partner	The Korean SME partner will play a central role in translating the developed modular retrofit platform into an industrially viable, market-oriented solution. Its responsibilities focus on manufacturing implementation, validation under real-world production conditions, and preparation for commercialization within the Korean mobility ecosystem. Specifically, the Korean SME partner is expected to: • Refine, adapt, and industrialize the modular platform design with a focus on small-series and variant-oriented production. • Perform prototype manufacturing, assembly, and system integration in accordance with the defined modular architecture and retrofit processes. • Conduct functional testing, validation, and quality assurance activities from an SME production and operational perspective. • Optimize manufacturability, assembly workflows, and cost structures to ensure scalability, efficiency, and economic feasibility. • Contribute to commercialization planning, including market adaptation, customer requirements, and regulatory considerations within the Korean context. • Provide structured feedback for iterative improvement of modular design, drivetrain integration, and ergonomic configuration based on hands-on manufacturing and testing experience. • Participate in knowledge exchange and training activities to ensure workforce readiness and effective implementation of the platform within SME environments.

	Through these contributions, the Korean SME partner ensures that the jointly developed technology moves beyond conceptual validation toward practical manufacturability, market relevance, and sustainable commercialization, strengthening the competitiveness of Korean SMEs in advanced and circular mobility markets.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes of this Joint R&D project go beyond the realization of a single technical demonstrator and focus on establishing a scalable, industrially relevant technology platform for sustainable urban mobility. From a technological perspective, the project will deliver a validated modular retrofit platform for the electrification and sustainable reuse of motorcycles. Key results include a clearly defined system architecture, standardized mechanical and electrical interfaces, and a repeatable retrofit and integration process. The project will demonstrate that retrofit-based electrification can meet performance, safety, and sustainability requirements while remaining compatible with SME manufacturing environments and small-series production. By embedding regulatory awareness and compliance-oriented design principles from an early development stage, the project is expected to significantly reduce transition barriers from prototype to market-ready solutions. This approach supports accelerated validation, lowers approval-related risks, and enables SME-driven commercialization of modular electric motorcycles for urban mobility applications. In terms of commercialization potential, the project enables new and flexible business models for Korean SMEs, including retrofit services for existing vehicles, small-series electric motorcycle production, and modular component supply. Compared to conventional full-vehicle development, the proposed platform lowers development effort and upfront investment, allowing SMEs to enter electric mobility markets with reduced risk and shorter time-to-market. The modular architecture further supports product diversification and customer-specific adaptation. From an industrial and economic perspective, the project contributes to strengthening Korea's SME ecosystem in advanced and sustainable mobility. The combination of international system-level expertise with Korean manufacturing, testing, and commercialization capabilities enhances technological competitiveness and fosters long-term industrial collaboration. In addition, the project supports workforce skill development in key areas such as electric drivetrain integration, modular system engineering, and sustainable manufacturing processes. From a societal and environmental perspective, the project promotes reduced resource consumption and lower CO_2 emissions by extending product lifecycles and enabling the systematic reuse of high-value components. This contributes to circular economy objectives and supports the transition toward more space-efficient, locally emission-free urban mobility solutions. Overall, the project is expected to generate lasting technological, economic, and environmental impact, positioning Korean SMEs as active contributors to emerging markets for electric, modular, and circular urban mobility solutions.

6. References

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(4) **Public Prototype Demonstration:** *Modular Electric Motorcycle Retrofit Prototype – Public Demonstration at Glemseck 101.*Glemseck 101 Motorcycle Festival, Leonberg, Germany, September 2025. Public event demonstrating the functional prototype, modular architecture, and real-world usability of the retrofit concept.
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(6) **Technical Validation and Sustainability Assessment:** *Technical Validation Report and Lifecycle Assessment of a Modular Electric Motorcycle Retrofit Platform:* Internal project documentation, Hochschule Esslingen, Labor für Fahrzeug- und Karosserieentwurf (LFK), 2025. Includes functional testing of 48V electric drivetrain integration and quantitative sustainability assessment; available to funding bodies upon request.